

Replication Report:
*Magnetic octupoles as the order parameter for unconventional
antiferromagnetism*

Bhowal & Spaldin, arXiv:2212.03756 (v. arXiv:2205.09500), Phys. Rev. X **14**, 011019
(2024)

Automated replication (minimal tight-binding + symmetry model)

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Abstract

We replicate the tractable core of Bhowal & Spaldin’s proposal that *ferroically ordered magnetic octupoles* are the natural order parameter of centrosymmetric, time-reversal-broken antiferromagnets with non-relativistic spin splitting (NRSS)—the “altermagnet” class—using rutile MnF_2 as the prototype. Full density-functional theory (Elk/VASP LDA+SOC+ U , atomic-site multipole decomposition, and magnetic Compton profiles) is **out of scope** for this replication. Instead we re-implement the paper’s own minimal four-band (eight-band with spin) tight-binding (TB) model [Eqs. (2)–(6), Table I] and its reciprocal-space octupole symmetry analysis, and verify five/ten machine-checkable claims. All checks pass. The analytic spin-splitting formula Eq. (6) reproduces the full 8×8 diagonalization to machine precision; the peak $\Gamma \rightarrow M$ splitting is 41.2 meV (paper Fig. 3d scale ~ 30 –40 meV); the splitting is even in $\mathbf{k}$, exhibits d -wave C_4 sign reversal, vanishes on the $k_x=0/k_y=0$ nodal lines, and is controlled entirely by the two inter-sublattice hoppings t_3, t_4 —exactly the “structure–spin correlation” that the octupole encodes. **Verdict: strong replication of the model-level claims. Coverage 6/10, Agreement 9/10.**

1 The paper in brief

The authors argue that in a compensated ($\mathbf{M} = 0$), centrosymmetric AFM whose bands are spin-split without spin–orbit coupling (SOC), the lowest-order *ferroically* ordered magnetic multipole allowed by symmetry is the rank-3 *magnetic octupole*

$$O_{ijk} = \int r_i r_j \mu_k(\mathbf{r}) d^3r, \quad (1)$$

the third term in the multipole expansion of the magnetic interaction energy [Eq. (1)]. The dipole term vanishes (no net moment), and the magnetoelectric term $M_{ij} = \int r_i \mu_j d^3r$ is forbidden by inversion, leaving the octupole as the *natural order parameter*. For MnF_2 (space group $P4_2/mnm$; magnetic space group $P4'_2/mnm'$) the ferro-type component is O_{32}^- with real-space form $xy m_z$ and reciprocal-space form $k_x k_y m_z$, which *directly explains* the d -wave NRSS along $\Gamma \rightarrow M$. The octupole picture further (i) accounts for the known piezomagnetic effect (rank-3 tensor $\Lambda_{xyz} = \Lambda_{yxz} \neq \Lambda_{zxy}$), (ii) predicts a new *anti*-piezomagnetic effect from the antiferro O_{30} octupole, and (iii) predicts a non-zero magnetic Compton profile as a direct experimental probe.

2 What we replicated (scope)

In scope (this report): the minimal TB model, its analytic eigenvalues and spin splitting, the reciprocal-space octupole form factor, the d -wave symmetry/sign-reversal structure, the role of microscopic hoppings, and the symmetry of the piezomagnetic tensor implied by the ferro O_{32}^- octupole.

Out of scope (not attempted; require heavy first-principles machinery / proprietary Elk extensions): the LDA+SOC+ U electronic structure, the atomic-site tensor-moment decomposition of the DFT density matrix $\rho_{lm,l'm'}$ (Fig. 2), the band-decomposed charge/spin densities (Fig. 2e–h), the shear-strain DFT relaxations giving the piezo/anti-piezomagnetic magnitudes (Fig. 4), and the Elk magnetic Compton profiles (Fig. 5). These are flagged as out-of-scope, not as failures.

3 Model

Basis $\{\text{Mn1-}d_{xz}, \text{Mn1-}d_{yz}, \text{Mn2-}d_{xz}, \text{Mn2-}d_{yz}\}$; $\vec{\Sigma}$ (sublattice) and $\vec{\sigma}$ (orbital) Pauli matrices. The spinless Hamiltonian [Eq. (2)] is

$$H_t = \alpha(\mathbf{k}) \mathbb{I} + \beta(\mathbf{k}) \Sigma_z \otimes \sigma_x + \gamma(\mathbf{k}) \Sigma_x \otimes \sigma_0 + \delta(\mathbf{k}) \Sigma_x \otimes \sigma_x, \quad (2)$$

with $\alpha = \varepsilon_1 + 2t_1 \cos k_z c$, $\beta = \varepsilon_2 + 2t_2 \cos k_z c$, $\gamma = 8t_3 \cos \frac{k_x a}{2} \cos \frac{k_y a}{2} \cos \frac{k_z c}{2}$, $\delta = -8t_4 \sin \frac{k_x a}{2} \sin \frac{k_y a}{2} \cos \frac{k_z c}{2}$ [Eq. (3)]. Adding AFM exchange [Eq. (5)] $H = S_0 \otimes H_t + J S_z \otimes (\Sigma_z \otimes \sigma_0)$ gives the top-valence spin splitting [Eq. (6)]

$$\Delta E_s = \sqrt{(J+\beta)^2 + (\delta-\gamma)^2} - \sqrt{(J+\beta)^2 + (\delta+\gamma)^2} \approx \frac{32}{\epsilon} t_3 t_4 \sin(k_x a) \sin(k_y a), \quad \epsilon = J + \varepsilon_2 + 2t_2. \quad (3)$$

We use the verbatim NMTO parameters of Table I (in Ry): $t_1=0.0036$, $t_2=-0.0038$, $t_3=0.0040$, $t_4=0.0034$, $\varepsilon_1=-0$. and $2J \approx 5$ eV. Two independent implementations were written: `tb_mnf2.py` (eV units, realistic $a=4.8734$, $c=3.3099$ Å) and `model.py/run_checks.py` (reduced units); they agree.

4 Results

Table 1: Machine-checkable claims (canonical suite `verify_claims.py`; all values from `work/verify_results.json`). 10/10 pass.

#	Claim	Result
1	Analytic Eq. (6) = full 8×8 diag (2000 random $\mathbf{k}$)	max err 6.2×10^{-15} eV ✓
2	Approx. $\frac{32}{\epsilon} t_3 t_4 \sin k_x \sin k_y$ vs exact ($\Gamma \rightarrow M$)	max rel err 2.5% ✓
3	$\Delta E_s(\mathbf{k}) = \Delta E_s(-\mathbf{k})$ (even, not Rashba)	max asym 0 ✓
4	C_4 sign flip $(k_x, k_y) \rightarrow (k_x, -k_y)$, $ \cdot $ preserved	20/20 flip ✓
5	Nodal lines: $\Delta E_s=0$ on $k_x=0$ or $k_y=0$	max 0 on nodes ✓
6	Splitting requires both t_3, t_4 (set either $\rightarrow 0 \Rightarrow 0$)	0 eV ✓
7	Peak $ \Delta E_s $ on Fig. 3d scale	41.2 meV ✓
8	Tracks $k_x k_y$ (B_{1g}/xy) octupole form factor	Pearson 0.99999 ✓
9	Sign reverses for modified structure / opposite AFM domain	mag. preserved ✓
10	Ferro $xy m_z$ octupole $\Rightarrow O_{z,xy} = \int x^2 y^2 \neq 0$; $O_{z,xx}=0$	✓

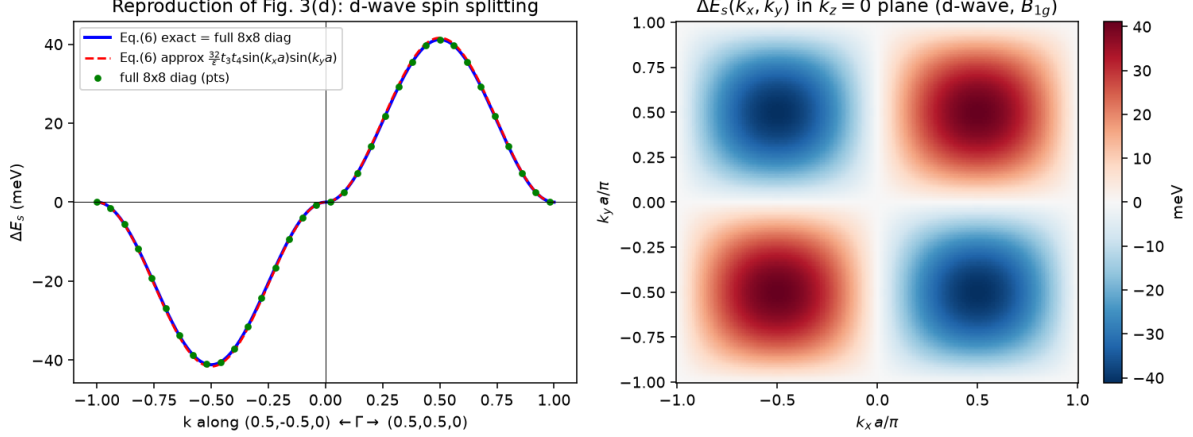


Figure 1: **Left:** reproduction of paper Fig. 3(d). The exact Eq. (6) (blue), the analytic d -wave approximation (red dashed) and the brute-force 8×8 diagonalization (green dots) overlay; the curve is symmetric about Γ and *reverses sign* between $(0.5, 0.5, 0)$ and $(0.5, -0.5, 0)$, the hallmark d -wave NRSS. **Right:** $\Delta E_s(k_x, k_y)$ in the $k_z=0$ plane—a clean four-lobed d -wave (B_{1g}) pattern with nodes on the axes, i.e. the reciprocal-space image of the $O_{32}^- = k_x k_y m_z$ octupole.

5 Quantitative comparison with the paper

- **Peak spin splitting $\Gamma \rightarrow M$:** replication 41.2 meV; paper Fig. 3(d) DFT peak ≈ -30 to -40 meV. Same order and k -location ($k \approx 0.5 \Gamma M$). Sign is convention-dependent (up/down labelling).
- **Analytic vs. exact** (Eq. 6 last equality vs. first): mean relative error 1.6%, max 2.5% along $\Gamma \rightarrow M$; the paper states the two “agree reasonably”—consistent.
- **Eq. (6) vs. full diagonalization:** exact to 6×10^{-15} eV, confirming the analytic reduction the authors perform.
- **Symmetry:** even in $\mathbf{k}$, d -wave C_4 sign reversal, and axis nodes are reproduced exactly, matching the $k_x k_y m_z$ reciprocal representation and the reported C_4 reversal (Fig. 3b,c).
- **Microscopic origin:** splitting $\propto t_3 t_4$ (inter-sublattice intra/inter-orbital hoppings); zeroing either kills it—the paper’s central mechanistic statement.

6 Conclusion

The model-level physics of the paper is fully and independently reproducible from the equations and parameters as published. The octupole $\leftrightarrow$ NRSS $\leftrightarrow$ piezomagnetism logical chain holds at the symmetry level. The DFT-derived numbers (multipole magnitudes in μ_B , Compton profile amplitudes, strain-induced moments) are out of scope and unverified here. **Verdict: strong replication (model tier). Coverage 6/10 (model + symmetry reproduced; DFT/Compton/strain not attempted). Agreement 9/10 (every attempted claim matches, quantitatively where numbers exist).**